

Learning Objective 1.3: I can determine the polarization of an antenna and describe the bandwidth characteristics of common antenna types.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Identify the polarization.** For each plane wave propagating in $+\hat{\mathbf{z}}$, identify the polarization (linear, RHCP, LHCP, or elliptical) and — where linear — the tilt angle relative to $\hat{\mathbf{x}}$.

(a) $\vec{E} = \hat{\mathbf{x}}3\cos(\omega t - kz)$

(b) $\vec{E} = \hat{\mathbf{x}}2\cos(\omega t - kz) + \hat{\mathbf{y}}2\cos(\omega t - kz)$

(c) $\vec{E} = \hat{\mathbf{x}}5\cos(\omega t - kz) + \hat{\mathbf{y}}5\cos(\omega t - kz - 90^\circ)$

(d) $\vec{E} = \hat{\mathbf{x}}5\cos(\omega t - kz) + \hat{\mathbf{y}}5\cos(\omega t - kz + 90^\circ)$

(e) $\vec{E} = \hat{\mathbf{x}}3\cos(\omega t - kz) + \hat{\mathbf{y}}1\cos(\omega t - kz - 90^\circ)$

2. **Axial ratio.** A satellite downlink antenna is spec'd circularly polarized with $\text{AR} \leq 3 \text{ dB}$ across its operating band.

- (a) Convert the 3 dB axial ratio to a linear ratio (major/minor axis).

- (b) Write the two orthogonal linear components (up to a common scale) with the appropriate 90° phase, assuming the ellipse major axis is along $\hat{\mathbf{x}}$.

- (c) An $\hat{\mathbf{x}}$ receiver picks off E_x ; a $\hat{\mathbf{y}}$ receiver picks off E_y . Compute the ratio of received *power* in dB. Should it match part (a)?

3. Polarization loss factor.

- (a) A ground station transmits a *linearly* polarized signal along $\hat{\mathbf{x}}$. A satellite receiver is **RHCP** and pointed correctly. Compute the PLF in dB.

- (b) Same ground station, but the satellite receiver is **LHCP**. Compute the PLF.

- (c) Two linear whip antennas: radio A vertical, radio B tilted 30° from vertical. Compute the PLF.

- (d) In one sentence, explain why aircraft-to-ground links almost always specify circular polarization on at least one end.

4. **Impedance bandwidth from a VSWR spec.** An antenna is spec'd “VSWR ≤ 2 from 2.30GHz to 2.55GHz.”

(a) What is the impedance bandwidth in MHz?

(b) What is the fractional bandwidth (FBW) in percent?

(c) Convert VSWR = 2 to $|\Gamma|$ and to return loss in dB.

(d) Narrowband, broadband, or ultra-wideband? Likely family — patch, horn, or log-periodic?

5. Fractional and ratio bandwidth.

- [illegible]

6. **Chu-Harrington intuition.** A design must fit inside a sphere of radius $a = 15$ mm and operate at $f = 915$ MHz (ISM band).

(a) Compute λ , k , and the electrical size ka .

(b) Estimate the minimum Q from $Q_{\min} \approx \frac{1}{(ka)^3} + \frac{1}{ka}$.

(c) Estimate the maximum fractional bandwidth $\text{FBW}_{\max} \approx 1/Q_{\min}$ (for $\text{VSWR} \leq 2$).

(d) The designer wants 100 MHz of usable bandwidth at 915 MHz. Possible in a 15 mm sphere?

7. **Reading the datasheet.** A vendor offers a GPS patch specified as **RHCP**, $AR \leq 3$ dB , “bandwidth 4% ($VSWR \leq 2$)”. You are choosing the antenna for the other end of the link.

(a) The datasheet defines handedness *from the receiver looking back toward the incoming wave*. Which convention is that, what is the part’s sense in **IEEE** terms, and what goes wrong if you pair it with a satellite transmitting IEEE RHCP?

(b) Convert $AR = 3$ dB to a linear ratio A , and give A^2 .

(c) A *linear* antenna receives from this patch. Using $PLF(\theta) = \frac{A^2 \cos^2 \theta + \sin^2 \theta}{A^2 + 1}$, find the best and worst PLF over all orientations θ , in dB.

(d) Show that the peak-to-null swing in dB equals the axial ratio in dB.

(e) The far end is an opposite-sense CP antenna, also $AR = 3$ dB , major axes aligned. Worst-case rejection is $\left(\frac{A^2 - 1}{A^2 + 1} \right)^2$. Evaluate it and compare with the idealized PLF table.

- (f) Name two things you must ask before believing the antenna works over the advertised 4%.

Documentation: _____